# Transformer-Based Belief Modeling for Offline Policy Learning in Partially Observed, Simultaneous-Action Games

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**Abstract**

Many real decision problems are partially observed, require simultaneous action commitment, and exhibit combinatorial structure that causes naive learners to memorize frequent patterns rather than infer hidden state. This thesis studies whether causal transformer sequence models can maintain useful beliefs over hidden opponent attributes from interaction history and whether explicit belief supervision improves offline policy learning under combinatorial distribution shift.

We propose a hierarchical tokenization scheme with within-turn self-attention and cross-turn causal attention, auxiliary belief prediction heads over active-opponent attributes, and a combinatorial evaluation protocol based on held-out opponent team keys. Experiments use Metamon Gen 9 OU human replays. Competitive Pokémon is the benchmark domain only. The scientific target is belief inference and generalization under offline imitation, not ladder win rate or search-based play strength.

On a 2 000-battle belief-enriched corpus, turn-only behavioral cloning reaches 20.4% top-1 action match on random battle holdout and 45.1% on held-out team keys. Explicit belief heads reach 28.1% move-slot accuracy and beat a GRU baseline, while a latest-turn MLP reaches 35.5%. Auxiliary belief and matchup modules do not improve action match or team-level generalization over turn-only cloning. The opponent-team generalization gap collapses with data scale for every architecture. These results support nontrivial offline belief learning and show that belief-to-policy transfer is not automatic at Master's-scale models and data.

**Keywords:** partial observability, offline reinforcement learning, transformers, opponent modeling, combinatorial generalization, behavioral cloning

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# Chapter 1

# Introduction

## 1.1 Motivation

Many decision problems in security, negotiation, finance, and multi-agent interaction share three properties that standard fully observed reinforcement learning benchmarks underrepresent.

**Partial observability.** Important state variables remain hidden until late in an interaction. An agent must infer latent structure from noisy clues rather than read full state directly.

**Simultaneous action selection.** Both sides commit to actions before the environment resolves the step. This differs from chess-like turn-taking and creates a distinct learning problem around anticipating opponent intent under uncertainty.

**Combinatorial structure.** The space of valid configurations grows quickly. A model that memorizes frequent patterns may appear strong on average yet fail on rare but valid combinations held out from training.

Transformers are a natural fit for these settings because they aggregate evidence across long interaction histories with causal attention. Offline learning is appropriate when high-quality human trajectories are available at scale and the deployment goal is a single forward pass without lookahead search or simulator queries at inference time.

## 1.2 Problem Statement

Given a sequence of first-person observations from past turns, we seek a model that

(i) maintains a useful belief over hidden opponent attributes,

(ii) uses that belief to choose strong actions in new interactions, and

(iii) generalizes to opponent configurations outside the training distribution.

Formally, each turn produces an observation $o_t$ derived from public state. Hidden variables $z_t$ include unrevealed moves, items, abilities, and other private attributes. Actions $a_t$ are chosen under simultaneous-move constraints. The model receives offline trajectories $\{(o_{1:T}, a_{1:T}, z_{1:T})\}$ reconstructed from full logs where $z$ is known post hoc but not visible to the agent at decision time.

The central thesis question is whether explicit belief supervision and transformer-based history modeling improve both belief accuracy and policy performance under combinatorial distribution shift.

## 1.3 Scope and Non-Goals

This thesis measures offline imitation and belief metrics. It does *not* claim that the resulting agent is competitive against human ladder players. Live win rate is a separate play-strength question. Offline action match of roughly twenty percent on hundreds of classes is far above chance and still leaves most turns incorrect, so weak live performance against skilled humans is expected and does not invalidate the scientific measurements below.

Pokémon is used only as a benchmark. The contribution is an empirical study of belief inference and combinatorial generalization under offline learning, not a product bot or a search-based competitive agent.

## 1.4 Research Questions

**RQ1 (Belief inference)** Can a causal transformer predict hidden opponent attributes more accurately than classical and neural baselines?

**RQ2 (Belief usefulness)** Does improved belief accuracy translate into better action selection under behavioral cloning and offline policy metrics?

**RQ3 (Combinatorial generalization)** Under held-out team splits, do belief-augmented policies degrade more gracefully than action-only models?

**RQ4 (Architecture)** How much gain comes from within-turn structure, cross-turn causal attention, and auxiliary belief heads versus simpler baselines?

## 1.5 Contributions

This thesis makes four contributions.

1. A structured tokenization and belief supervision framework for partially observed simultaneous-action games.
2. Empirical evidence that transformers and classical baselines learn nontrivial beliefs over hidden opponent moves from offline human play, and that a latest-turn MLP can outperform a history transformer on that target.
3. A combinatorial generalization protocol with held-out opponent team keys, showing that team-level OOD gaps shrink primarily with data scale rather than with explicit matchup or belief modules.
4. An open-source implementation built on Metamon, poke-env, and Gymnasium compatible tooling, including export, training, probing, and OOD evaluation scripts.

## 1.6 Evaluation Domain

We evaluate on competitive Pokémon battles reconstructed from human spectator replays via the Metamon dataset [Grigsby et al., 2023, 2025]. Pokémon combines partial observability, simultaneous move commitment, and combinatorial team structure in one domain with large-scale offline data. Gen 9 OU team preview reveals all opponent species at battle start, so species belief is empty by design and active-opponent move, item, ability, and tera labels are the primary belief targets. The methods apply to any domain with similar structure.

## 1.7 Thesis Organization

Chapter 2 reviews POMDPs, transformers in decision making, and offline RL. Chapter 3 briefly describes the benchmark environment. Chapters 4 and 5 present tokenization and the belief transformer. Chapters 6 to 8 cover the experiment plan, results, and interpretability analysis. Chapters 9 and 10 discuss limitations and future work.

# Chapter 2

# Background

## 2.1 Partially Observable Decision Making

A partially observable Markov decision process (POMDP) extends the MDP formalism with a latent state $s_t$ that is not fully observed [Kaelbling et al., 1998]. The agent receives observations $o_t$ and must maintain a belief $b_t(s) = P(s_t = s \mid o_{1:t}, a_{1:t-1})$ over latent state. Optimal control in POMDPs requires planning over belief space, which is often intractable in large discrete domains.

Recent work amortizes belief inference with neural sequence models that map interaction history to action distributions or latent embeddings [Chen et al., 2021, Liu et al., 2024]. This thesis asks whether structured auxiliary supervision over interpretable hidden attributes improves both belief quality and downstream policy robustness.

## 2.2 Simultaneous-Action Games

In simultaneous-move settings both agents select actions before the environment advances. The effective decision problem is a partially observed stochastic game. Unlike turn-based games, the agent cannot observe the opponent's current action before committing. Policies must therefore encode predictions about opponent intent from history alone.

## 2.3 Transformers for Sequential Decision Making

The transformer architecture uses self-attention to relate tokens within a sequence [Vaswani et al., 2017]. Decision Transformer and related methods apply causal transformers to offline RL by modeling return-conditioned action sequences [Chen et al., 2021]. In competitive settings, opponent modeling methods such as TAO embed opponent trajectories for in-context adaptation [Liu et al., 2024]. The Self-Confirming Transformer generates fictitious belief tokens representing inferred opponent actions and feeds them back into policy decoding [Zhang et al., 2024].

Our approach differs in three ways. We predict structured hidden attributes per opponent slot rather than only next-opponent-action embeddings. We use a hierarchical within-turn and cross-turn encoder rather than a flat token stream. We evaluate combinatorial held-out team and set splits as a primary metric rather than leaderboard rank alone.

## 2.4 Offline Reinforcement Learning

Offline RL learns from a fixed dataset without online environment interaction [Sutton and Barto, 2018]. Behavioral cloning (BC) imitates expert actions and provides a stable initialization but suffers from covariate shift. Conservative Q-Learning (CQL) penalizes overestimation on out-of-distribution actions [Kumar et al., 2020]. AMAGO provides another sequence-model fine-tuning path compatible with transformer backbones.

We use BC with auxiliary belief losses as Stage A and optional CQL or AMAGO fine-tuning as Stage B. Inference remains a single forward pass with no search.

## 2.5 Combinatorial Generalization

Humans generalize to novel combinations of known primitives [Lake et al., 2018]. Machine learning systems often fail when test combinations were absent from training even when individual primitives were seen. We adopt held-out team keys and held-out move-set tuples as explicit tests of combinatorial generalization rather than random battle splits alone.

## 2.6 Positioning Relative to Metamon

Grigsby et al. [2025] demonstrated that large sequence models trained on human Pokémon replays can reach strong play without search. That work establishes feasibility of transformer policies in this domain. Our thesis does not aim to beat their agents on ladder rank. Instead we study explicit belief inference, calibration, and OOD degradation under combinatorial splits, which their paper does not center.

# Chapter 3

# Benchmark Environment

## 3.1 Why Competitive Pokémon

Competitive Pokémon Singles (CPS) is a turn-based strategy game with simultaneous move selection each turn. Players choose teams of six Pokémon with moves, items, abilities, and in Generation 9 a terastallization type. At battle start in Gen 9 OU, team preview reveals all opponent species. Unrevealed moves, items, abilities, and terastallization type remain hidden and must be inferred from damage ranges, switch patterns, item triggers, and move reveals.

CPS therefore combines partial observability, simultaneous commitment, long horizons (often 20–100+ turns), and combinatorial team structure. Human replay data spans more than a decade and grows daily on public ladders.

## 3.2 Infrastructure

We use three open components.

- **poke-env** bridges Python agents to a local Pokemon Showdown server for live evaluation.
- **Metamon** reconstructs first-person trajectories from spectator replays with full-state annotations for belief labels [Grigsby et al., 2023].
- **Gymnasium** provides the environment interface standard for RL tooling in this project.

## 3.3 Scope of This Chapter

This thesis is not about Pokémon mechanics, tier lists, or heuristic engines. We use Gen 9 OU human battles as the primary corpus. Random-bot logs from poke-env support pipeline debugging only. All scientific claims concern belief inference and generalization, with CPS as an empirical testbed.

## 3.4 Action Space

At each decision point the agent may use a move targeting an opponent slot or switch to a bench Pokémon. Actions are encoded as discrete classes in an action vocabulary built from training data. The exact action cardinality depends on format rules and logged move names.

## 3.5 Data Pipeline Status

The implementation roadmap for this draft is complete for Stage A offline experiments.

1. Local Showdown server and baseline logging **(complete)**.
2. 13-token turn encoder and vocabulary **(complete)**.

3. TurnTransformer behavioral cloning **(complete)**.
4. Metamon bridge and belief-enriched export **(complete)**.
5. Multi-task belief training and matchup ablations **(complete)**.
6. Split A / Split B combinatorial evaluation harness **(complete)**.
7. Live poke-env win-rate appendix and Split C **(deferred)**.

Appendix A lists repository paths for each component.

# Chapter 4

# Representation and Belief Targets

## 4.1 Design Goals

The observation representation must satisfy four constraints.

1. Fixed structure each turn so batching and attention masks are simple.
2. Explicit separation of field state, player party, and opponent party.
3. Compatibility with first-person partial observations from Metamon.
4. Discrete tokens suitable for embedding lookup and vocabulary growth.

## 4.2 Thirteen-Token Turn Layout

Each turn is encoded as exactly thirteen discrete tokens.

Table 4.1: Turn token layout. Indices are zero-based.

| Index | Role |
|---|---|
| 0 | Field (weather, terrain, hazards on both sides) |
| 1–6 | Player party slots (padded to six) |
| 7–12 | Opponent party slots (partial info when unrevealed) |

The field token concatenates sorted effect keys for weather, terrain, and side conditions. Each party token encodes species, discretized HP bucket, status, active flag, and fainted flag. Unfilled slots use a padding token.

A battle is a sequence of turns. Training samples use a causal context window of $N$ prior turns plus the current turn. Left padding aligns the most recent turn to the final sequence position.

## 4.3 History Window

Given turn frames $x_1, \ldots, x_t$, we form context

$$C_t = \mathrm{pad}(x_{t-N+1}, \ldots, x_t)$$

with a binary turn mask marking valid versus padded positions. The model may not attend to future turns. This implements a finite-history approximation to belief updating in a POMDP.

## 4.4 Belief Supervision Targets

Belief heads predict hidden opponent attributes knowable from full logs but not from the first-person view at time $t$. Primary targets in priority order are

1. species identity for each unrevealed opponent slot,
2. moves for revealed or partially revealed opponent Pokémon,
3. held item when inferable before reveal,
4. ability when not yet shown,
5. terastallization type in Gen 9 OU.

Each target is a multi-class prediction per opponent slot. Revealed or empty slots are masked out of the loss. Labels are extracted from Metamon full-state annotations at each turn boundary.

For species belief we define a vocabulary over base species names. Slot $s \in \{0, \ldots, 5\}$ maps to opponent token index $7 + s$ and receives a label $y_{t,s} \in \mathcal{Y}_{\text{species}} \cup \{\emptyset\}$ where $\emptyset$ denotes masked slots.

## 4.5 Combinatorial Keys for Evaluation

We define two keys for distribution shift experiments.

**Team key** Sorted tuple of six species names for the opponent roster.

**Set key** Tuple of species, item, ability, and move quartet for a slot.

Held-out keys define test battles where the opponent configuration never appeared in training. This tests compositional generalization beyond random battle splits.

## 4.6 Implementation

The `TurnTokenizer` class in `src/pokemon_thesis/tokenizer/` implements encoding from JSON battle records. Encoded trajectories are stored as JSONL with `token_ids`, `action`, and optional `belief` fields for multi-task training.

# Chapter 5

# Model and Training

## 5.1 Overview

We extend the existing `TurnTransformer` backbone with auxiliary belief heads and optional belief fusion into the policy head. The design keeps parameter count modest (on the order of one to five million) for Master's-scale compute.

## 5.2 Hierarchical Encoder

### 5.2.1 Within-Turn Self-Attention

For each turn $t$, token embeddings are summed with slot position embeddings and passed through a transformer encoder over the thirteen slots:

$$h_t = \mathrm{MeanPool}(\mathrm{SlotEnc}(E(x_t) + P_{\mathrm{slot}})).$$

Mean pooling over slot outputs yields one turn embedding $h_t \in \mathbb{R}^d$.

### 5.2.2 Cross-Turn Causal Attention

Turn embeddings receive turn position embeddings and pass through a causal transformer encoder:

$$\tilde{h}_t = \mathrm{TurnEnc}_{\mathrm{causal}}(h_{1:t} + P_{\mathrm{turn}})_t.$$

The representation at the last valid turn index is used for prediction heads.

This two-level structure lets slot attention relate field state to individual party members within a turn while turn attention integrates evidence across time without peeking at future observations.

## 5.3 Policy Head

The policy head is a linear classifier over the final turn embedding:

$$\pi(a \mid C_t) = \mathrm{softmax}(W_\pi \tilde{h}_{t^*}),$$

where $t^*$ is the index of the most recent valid turn. Training minimizes cross-entropy against the human action (behavioral cloning).

## 5.4 Belief Heads

For each belief target type $k$ and opponent slot $s$, a small MLP maps $\tilde{h}_{t^*}$ to logits over the target vocabulary:

$$\hat{z}_{t,s}^{(k)} = \mathrm{softmax}\big(f_s^{(k)}(\tilde{h}_{t^*})\big).$$

Species prediction uses six slot-specific heads sharing the same trunk representation. Masked slots exclude terms from the belief loss.

## 5.5 Belief Fusion Variant

In an ablation we concatenate belief head softmax outputs to the policy input:

$$\pi(a \mid C_t) = \mathrm{softmax}\big(W_\pi[\tilde{h}_{t^*}; \hat{z}_t]\big).$$

This tests whether explicit belief features help action selection beyond shared representation learning in the trunk.

## 5.6 Training Procedure

### 5.6.1 Stage A: Belief-Augmented Behavioral Cloning

The multi-task loss is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_{\text{bel}} \sum_k \mathcal{L}_{\text{bel}}^{(k)}, \tag{5.1}$$

where $\mathcal{L}_{\text{action}}$ is action cross-entropy and each $\mathcal{L}_{\text{bel}}^{(k)}$ is masked cross-entropy over opponent slots. The weight $\lambda_{\text{bel}}$ is tuned on validation data.

### 5.6.2 Stage B: Conservative Offline RL Fine-Tuning

We initialize from Stage A and optionally fine-tune with CQL or AMAGO on the same offline trajectories. No search or simulator rollouts run at inference.

### 5.6.3 Stage C: Evaluation

All benchmark battles use frozen checkpoints. No online weight updates occur during evaluation.

## 5.7 Baselines

Belief baselines include gradient-boosted trees and logistic regression on hand-crafted features, MLP on the latest turn only, GRU over turn embeddings, TurnTransformer without belief loss, and a flat causal transformer over $13 \times T$ tokens.

Policy baselines include BC with MLP, BC with GRU, BC TurnTransformer, belief- augmented TurnTransformer, and belief-augmented models with offline RL fine-tuning.

## 5.8 Architecture Summary

The `BeliefTransformer` class implements belief heads and optional fusion. Training is invoked via `scripts/train_belief_bc.py`.

Table 5.1: Default hyperparameters for the belief transformer.

| Hyperparameter | Default |
|---|---|
| Model dimension $d$ | 128 |
| Attention heads | 4 |
| Slot encoder layers | 1 |
| Turn encoder layers | 2 |
| Context turns $N$ | 8 |
| Dropout | 0.1 |
| Optimizer | AdamW, lr $3 \times 10^{-4}$ |
| $\lambda_{\text{bel}}$ | tuned on validation |

# Chapter 6

# Experimental Setup

## 6.1 Dataset

The primary corpus is Metamon Gen 9 OU human replays. Phase 1 trains turn-only behavioral cloning on an encoded subset of approximately 255 000 turns. Phase 2 exports a belief-enriched corpus of 2 000 battles (69 411 turns) with active-opponent move, item, ability, and tera labels plus team matchup keys. Observations use first-person reconstruction from spectator logs. Actions come from human moves. Belief labels come from full-state opponent annotations that are unavailable to the agent at decision time.

Random-bot logs from poke-env support pipeline validation before human data is used for research runs.

## 6.2 Data Splits

We report two primary split strategies and defer a third.

### 6.2.1 Split A: Random Battle Split

Standard 80/20 train and validation partition by battle ID. This measures in-distribution imitation and belief accuracy on unseen battle trajectories.

### 6.2.2 Split B: Held-Out Team Cores

Each opponent team receives a canonical key from sorted species names. Teams in a designated held-out set never appear in training. Evaluation compares turns whose opponent team key appeared in training (ID train teams) against turns whose key was unseen (OOD teams). The generalization gap is ID train team accuracy minus OOD team accuracy and is the primary summary statistic for RQ3.

### 6.2.3 Split C: Held-Out Set Combinations (Deferred)

Holding out specific tuples of item, ability, and move quartet would test finer combinatorial shift beyond species identity. Split C is left for future work because Gen 9 OU already reveals species at team preview and because the scaled Phase 2 results already answer the species-key version of RQ3.

## 6.3 Belief Metrics

For move-slot belief we report top-1 accuracy on masked unrevealed slots. Baselines include a latest-turn MLP, a GRU over eight turns, and a linear probe on a frozen BC trunk. Item, ability, and tera heads are trained jointly with move heads. Aggregate move-slot accuracy is the primary RQ1 number reported in Chapter 7. Calibration plots (ECE, Brier) and reveal-turn anticipation curves are deferred.

## 6.4 Policy Metrics

Policy evaluation uses

- BC cross-entropy and top-1 action match versus human actions on Split A,
- Split B ID versus OOD team action match and the generalization gap,
- ablation comparisons among turn-only BC, matchup BC, ActiveBelief, and belief-fusion variants.

Live win rate on poke-env is an optional play-strength metric. It is not required to answer RQ1–RQ3 as framed in this thesis. Offline action match measures imitation fidelity, not ladder competitiveness against humans.

## 6.5 Ablations

Table 6.1: Ablation studies executed in Phase 2.

| Ablation | What it tests |
|---|---|
| Turn-only BC vs Matchup BC | Value of explicit team encoder |
| Matchup BC vs ActiveBelief | Value of auxiliary belief heads |
| Belief fusion vs parallel heads | Explicit belief features for policy |
| MLP / GRU / linear probe vs ActiveBelief | RQ1 baseline comparison |
| 500 vs 2 000 battle scale | Data coverage versus architecture |

Context-window sweeps, flat $13T$ sequence transformers, and offline RL (CQL or AMAGO) remain future work.

## 6.6 Statistical Reporting

We report point estimates on fixed validation and OOD splits shared across models. Paired comparisons use the same encoded corpus and token vocabulary. Bootstrap confidence intervals over battles are desirable for a journal revision but are not required for the Master's-scale tables in Chapter 7.

## 6.7 Minimum Viable Result

The thesis meets its pass bar if at least two of the following hold.

1. Belief models demonstrate nontrivial move-belief accuracy relative to chance and beat at least one strong neural baseline on that target.
2. Belief-augmented BC improves OOD team metrics or shrinks the generalization gap relative to turn-only BC.
3. Offline RL fine-tuning improves offline or live policy metrics over BC-only without search at inference.
4. Ablations show that structured turn history or scale contributes measurable signal beyond naive baselines (chance-level action match, empty belief labels, or pilot-scale OOD gaps).

As reported in Chapter 7, criteria (1) and (4) hold. Criterion (2) does not hold at current scale. Criterion (3) was not tested and is future work. Negative results on (2) are treated as scientific findings, not as thesis failure.

## 6.8 Compute and Reproducibility

Training runs on CPU or a single GPU. Checkpoints, vocabularies, and evaluation JSON files are saved under `checkpoints/` and `logs/`. The one-command Phase 2 pipeline is `scripts/scale_belief_research` The thesis PDF is built from `thesis/main.tex`.

# Chapter 7

# Results

## 7.1 Overview

This chapter reports behavioral cloning and combinatorial generalization results on Metamon Gen 9 OU human replays. Phase 1 established a turn-only `TurnTransformer` baseline on a large encoded subset. Phase 2 exported a belief-enriched corpus, trained three ablations on the same vocabulary, and evaluated them under random battle holdout (Split A) and held-out opponent team keys (Split B). All Phase 2 models use an eight-turn context window, $d = 128$, and twenty training epochs unless noted otherwise.

Corpus statistics for the scaled belief export are 69 411 turns from 2 000 battles, with 59 254 supervised move slots and thousands of item, ability, and terastallization labels on the active opponent. Gen 9 OU team preview reveals all opponent species at battle start, so species belief supervision is empty by design. Active-opponent move, item, ability, and tera type are the primary belief targets.

These tables measure offline imitation and belief accuracy. They do not report live win rate against human players. Weak live play against skilled humans is compatible with the offline numbers below and is outside the primary claim of this thesis.

## 7.2 Phase 1 Baseline (Full Subset)

The turn-only `TurnTransformer` was trained on `logs/encoded_metamon_subset.jsonl` (approximately 255 000 turns). On a random 20% validation split this model reached **18.4%** top-1 action match rate across 988 action classes (checkpoint `checkpoints/human_bc/bc_best.pt`). This exceeds the chance rate and confirms that the tokenization pipeline and hierarchical encoder learn non-trivial structure from human trajectories alone.

## 7.3 Policy Quality (RQ2)

### 7.3.1 Random Battle Holdout (Split A)

Table 7.1 reports validation action accuracy on the belief corpus after training on 80% of battles. All three Phase 2 models share the same encoded file and token vocabulary.

Turn history alone remains the strongest signal for action prediction at this scale. Explicit team matchup conditioning and auxiliary belief losses did not improve random-holdout action accuracy relative to the turn-only baseline. The best checkpoint for `active_belief` reached 20.2% validation action accuracy while simultaneously training move, item, ability, and tera heads.

Table 7.1: Top-1 action match on random validation battles (Split A), belief corpus, 2 000 battles, 20 epochs.

| Model | Action classes | Val. accuracy |
|---|---|---|
| TurnTransformer BC (`human_bc_belief`) | 425 | 20.4% |
| Matchup BC (`matchup_bc`) | 425 | 20.5% |
| ActiveBelief (`active_belief`) | 425 | 20.2% |
| ActiveBelief + fusion | 425 | 19.6% |

### 7.3.2 Live Evaluation

Live win-rate experiments on `poke-env` were not run for this draft. Offline action match of roughly twenty percent implies that most turns still disagree with the human demonstration. Against a skilled Gen 9 OU player, the agent would be expected to lose almost every game. That outcome would measure play strength, not whether belief learning or combinatorial evaluation succeeded. Live evaluation against `RandomPlayer`, frozen BC bots, and optionally ladder humans remains future work for a play-strength appendix.

## 7.4 Belief Inference (RQ1)

### 7.4.1 Active-Opponent Move Belief

The `ActiveBeliefTransformer` predicts four move slots plus item, ability, and tera type for the currently active opponent. Table 7.2 summarizes move-slot top-1 accuracy on the validation split.

Table 7.2: Move-slot belief accuracy on the validation split (2 000-battle corpus).

| Model | History | Val. move belief |
|---|---|---|
| Linear probe (frozen BC trunk) | 8 turns | 10.5% |
| GRU baseline | 8 turns | 17.2% |
| ActiveBelief transformer | 8 turns | 28.1% |
| **MLP baseline** | latest turn only | **35.5%** |

Scaling the export from 500 to 2 000 battles raised `active_belief` move belief from 15.5% to 28.1%. Classical baselines on the same labels show that a latest-turn MLP *outperforms* the transformer (35.5% versus 28.1%), while a GRU over eight turns reaches only 17.2%. Linear probes on the frozen BC trunk remain at 10.5% (Table 8.1). Much move information is therefore readable directly from the current turn tokens; deep causal history provides diminishing returns for this target at current scale.

### 7.4.2 Species and Set-Combination Beliefs

Species belief labels are unavailable in Gen 9 OU because team preview exposes all six opponent species before turn one. Item, ability, and tera heads were trained jointly with move heads, but this draft reports move-slot accuracy as the primary RQ1 metric because it has the densest supervision (59 254 labeled slots). Split C (held-out item, ability, and move-set tuples) and calibration metrics (ECE, Brier score) are deferred. The scientific claim for RQ1 is therefore limited to active-opponent move inference under the reported baselines.

## 7.5 Combinatorial Generalization (RQ3)

### 7.5.1 Held-Out Opponent Teams (Split B)

Each opponent team is keyed by sorted species names. Training battles cover 1 615 team keys. Evaluation compares turns from battles whose opponent team appeared in training (*ID train teams*) against turns from 385 held-out team keys never seen during training (*OOD teams*). The generalization gap is ID train team accuracy minus OOD team accuracy.

Table 7.3 shows results at pilot (500 battles) and scaled (2 000 battles) corpus sizes.

Table 7.3: Action match under held-out opponent team keys (Split B).

| Model | Scale | ID train | OOD teams | Gap |
|---|---|---|---|---|
| TurnTransformer BC | 500 | 43.3% | 41.4% | 1.9% |
| Matchup BC | 500 | 31.1% | 29.3% | 1.8% |
| TurnTransformer BC | 2 000 | 45.2% | 45.1% | 0.05% |
| Matchup BC | 2 000 | 42.6% | 42.1% | 0.4% |
| ActiveBelief | 2 000 | 44.3% | 44.4% | −0.03% |
| ActiveBelief + fusion | 2 000 | 43.8% | 43.3% | 0.5% |

On the pilot corpus, all models showed a one to two point gap between ID and OOD team splits. After scaling to 2 000 battles, the gap collapsed to near zero for every architecture. Turn-only BC retained the highest absolute OOD accuracy (45.1%). Matchup conditioning did not close the gap further and scored lower in absolute terms. ActiveBelief matched turn-only OOD performance despite multi-task belief training.

Random Split A accuracy (≈20%) is much lower than Split B team-conditioned accuracy (≈42–45%) because the splits measure different quantities. Split A holds out entire battle trajectories. Split B evaluates action match on turns grouped by whether the opponent roster was seen during training, which partially aligns train and test action distributions when species composition is the primary shift variable.

## 7.6 Architecture Ablations (RQ4)

Phase 2 implements three incremental ablations on a shared trunk.

- **TurnTransformer BC** uses only turn-history tokens.
- **Matchup BC** adds a `TeamMatchupEncoder` over both teams' six species and fuses the result into the final turn embedding before the action head.
- **ActiveBelief** extends Matchup BC with move, item, ability, and tera prediction heads trained with masked cross-entropy.

At 2 000 battles, matchup and belief modules did not outperform turn-only BC on either Split A or Split B. This suggests that, at current data and model scale, the thirteen-token turn representation already encodes most team and belief information useful for action matching. Explicit structure helps move inference (RQ1) more than policy metrics (RQ2).

### 7.6.1 Belief Fusion

We tested concatenating belief softmax outputs (four move slots, item, ability, tera) to the policy head input (Chapter 5). After twenty epochs, `active_belief_fusion` reached **19.6%** Split A validation action match and **27.6%** move belief, versus 20.2% and 28.1% for parallel heads without fusion. OOD team accuracy was 43.3% (fusion) versus 44.4% (without fusion). Feeding explicit belief probabilities into the action head did not improve imitation at this scale.

Flat $13T$ token transformers and context-window sweeps were not run. The executed ablations already show that adding matchup and belief modules on top of the hierarchical trunk does not improve policy metrics, while classical baselines clarify what history modeling contributes for move belief.

## 7.7 Stage B Offline RL

Conservative offline RL fine-tuning (CQL or AMAGO) initialized from belief- augmented BC was not run. Stage B is optional for thesis completion under the minimum viable bar in Chapter 6. Strong Stage A belief and BC results, including negative transfer findings, suffice for the claims made here.

## 7.8 Summary

Table 7.4 consolidates the scaled Phase 2 numbers. Belief accuracy refers to move-slot top-1 prediction for `active_belief` only. OOD action match is Split B OOD team accuracy.

Table 7.4: Results summary on the 2 000-battle belief corpus (20 epochs).

| Model | Move belief (val) | OOD action match | Gen. gap |
|---|---|---|---|
| TurnTransformer BC | — | 45.1% | 0.05% |
| Matchup BC | — | 42.1% | 0.4% |
| ActiveBelief | 28.1% | 44.4% | −0.03% |

### 7.8.1 Relation to Hypotheses

**H1 (belief inference).** Mixed support. The transformer learns non-trivial move beliefs and beats the GRU baseline (28.1% versus 17.2%), but a latest-turn MLP reaches 35.5% on the same split. History-conditioned attention is not required for strong move prediction when current-turn tokens already expose most clues.

**H2 (belief usefulness).** Not supported at current scale. Auxiliary belief loss did not improve action match over turn-only BC on either split.

**H3 (combinatorial generalization).** Mixed. Scaling data removed the OOD team gap for all models. Belief and matchup modules did not produce a smaller gap than the baseline.

**H4 (offline RL).** Not tested. Left as future work.

Reproducibility artifacts are stored under `checkpoints/` and `logs/ood_comparison.json`. The one-command replication script is `scripts/scale_belief_research.py`.

# Chapter 8

# Analysis and Interpretability

## 8.1 Overview

This chapter interprets the Phase 1 and Phase 2 results reported in Chapter 7. The analysis focuses on three patterns that emerged from scaled experiments on Metamon Gen 9 OU human replays

1. move-belief accuracy improves with data scale while policy metrics plateau across architectures,
2. opponent-team generalization gaps shrink with corpus size for every model, and
3. turn-only behavioral cloning matches or exceeds matchup- and belief-augmented variants on action prediction.

## 8.2 What the Thesis Shows Versus What It Does Not

The offline metrics support a scientific claim about learning, not a product claim about play. Nontrivial move belief and far-above-chance action match show that the representation and training pipeline recover structure from human logs. They do not imply that the agent would win live battles against real players. A model that matches the human action on roughly one turn in five still errs on most turns, and a single bad switch often decides a game. Interpreting the thesis as a failed competitive bot would misread the research questions.

The negative result on belief-to-policy transfer is equally important. Explicit belief heads improve decodable move prediction without improving action match. That finding warns against assuming that auxiliary opponent modeling will automatically improve imitation policies.

## 8.3 Scaling Belief Versus Scaling Policy

The `ActiveBeliefTransformer` increased validation move-slot accuracy from 15.5% on a 500-battle pilot export to 28.1% on a 2 000-battle export, while validation action accuracy rose from 8.5% to 20.2% over the same period. The belief head therefore benefits directly from additional labeled trajectories.

Action accuracy for all three Phase 2 models converged near 20% on random battle holdout (Split A) after scaling, with turn-only BC slightly ahead. Auxiliary belief losses add parameters and optimization pressure but did not produce a policy advantage at $d = 128$ and twenty epochs. Two explanations are consistent with the data.

**Shared trunk sufficiency.** The hierarchical turn encoder may already compress history into an embedding useful for both action and hidden-set prediction. Explicit belief heads learn to *decode* that embedding for supervised targets without changing the trunk enough to improve actions.

**Task interference.** Multi-task training divides gradient signal among move, item, ability, and tera heads. If $\lambda_{\text{bel}}$ is not carefully tuned, belief terms may slow action learning without improving generalization. Belief fusion into the policy input was tested and did not help (Chapter 7), so the remaining open knob is a careful $\lambda_{\text{bel}}$ sweep rather than fusion alone.

## 8.4 Why the Team Generalization Gap Collapsed

On the 500-battle corpus, Split B showed a 1.8–1.9 point gap between ID train teams and OOD teams. At 2 000 battles the gap fell below half a point for every architecture, including turn-only BC.

This pattern suggests that the pilot gap was driven largely by *data coverage* rather than architectural failure. With only 500 battles, many held-out team keys had few or no nearby examples in embedding space. Expanding to 2 000 battles increased overlap in species co-occurrence patterns and move distributions, so OOD team turns became easier to match even without explicit matchup modules.

Matchup BC did not outperform turn-only BC on Split B at scale. The `TeamMatchupEncoder` pools six species embeddings per side and adds them to the final turn vector. Much of that information is already present in the opponent party tokens within each turn's thirteen-token block. The explicit module may therefore be redundant given current tokenization.

Split B accuracy (≈42–45%) exceeds Split A accuracy (≈20%) because the metrics condition on different splits. Split A withholds entire battles. Split B groups turns by roster key while many action contexts remain familiar. Reporting both is essential to avoid over-interpreting high team-split numbers as solved imitation.

## 8.5 Linking Beliefs to Actions

The original plan correlated per-battle belief accuracy with per-battle action match on held-out teams. Species belief was unavailable in Gen 9 OU because team preview reveals all opponent species. Move-belief correlation analysis is still informative.

At scaled training, `active_belief` reached 28.1% move belief and 44.4% OOD team action match, while turn-only BC reached 45.1% OOD action match with no belief head. Higher move-belief accuracy therefore did not coincide with better OOD policy performance in this run.

Linear probes on frozen `human_bc_belief` trunk embeddings tell a sharper story (Table 8.1, script `scripts/probe_trunk_belief.py`). A fifteen-epoch linear probe on the final turn embedding reached only **10.5%** validation move-slot accuracy, versus **27.7%** for `active_belief` move heads on the same split. The BC trunk therefore does *not* expose hidden moves through an easy linear readout. Explicit belief heads and multi-task training learn representations that decode move labels far better than post-hoc probing.

Action accuracy nevertheless remains similar across models. Belief-specific features appear to live in a subspace that the shared action head does not exploit, or multi-task training reshapes the trunk in ways that help belief decoding without improving the action logit channel used at inference.

Table 8.1: Move-slot belief accuracy on validation battles (same split).

| Method | Val. move belief |
|---|---|
| Linear probe on frozen BC trunk | 10.5% |
| ActiveBelief move heads | 27.7% |

## 8.6 Implicit Belief in the Turn-Only Trunk

Turn-only BC and ActiveBelief achieved nearly identical OOD team accuracy (45.1% versus 44.4%) despite a large difference in supervised move-belief accuracy. Linear probes on the frozen BC trunk reached only 10.5% move belief (Table 8.1), so implicit linear encodings of unrevealed moves are weak. Explicit belief heads add substantial decodable structure without translating into better action match at current scale.

The thirteen-token layout exposes opponent party state, field conditions, and revealed moves each turn. Causal attention over eight turns integrates switch patterns, damage clues, and item triggers. That design aligns with the thesis motivation for structured partial observability without a separate belief filter.

## 8.7 Gen 9 OU and Belief Target Selection

Team preview fundamentally changed the belief task relative to older formats where species were hidden. The implementation pivot to active-opponent move, item, ability, and tera labels was necessary for non-empty supervision.

Species-centric hypotheses (H1 as originally framed) do not apply verbatim to this benchmark slice. Move-belief results should be read as evidence that history- conditioned transformers and classical baselines infer unrevealed moves, not that the full hidden-state posterior is solved.

## 8.8 Interpretability Directions Not Executed

Shallow decision trees on hand-crafted turn features and attention visualizations were planned to compare interpretable clues against neural errors. Those studies were not completed for this draft. Expected error modes remain relevant for future case studies

- rare move sets on held-out teams,
- ambiguous damage ranges before stat clues accumulate,
- deliberate unconventional sets underrepresented in training,
- late-game turns with truncated history relative to the eight-turn window.

The quantitative baselines already provide the main interpretability result for RQ1. A latest-turn MLP beating the transformer suggests that most move-belief signal lives in the current observation rather than in long-horizon attention patterns. Attention case studies would refine that story but are not required to support the claim.

## 8.9 Takeaways for RQ1–RQ4

**RQ1.** Transformers beat GRU on move belief but not a latest-turn MLP, which reaches 35.5% validation accuracy versus 28.1% for `active_belief`.

**RQ2.** Belief augmentation did not improve action match over turn-only BC at current scale. Offline imitation success does not imply live competitiveness against humans.

**RQ3.** Combinatorial team shift was mitigated primarily by scaling data, not by matchup or belief modules.

**RQ4.** Hierarchical turn structure appears sufficient for strong BC at this scale. Belief fusion into the policy head did not improve action match over parallel heads alone (19.6% versus 20.2% Split A). Flat-sequence and context-window sweeps remain open.

These interpretations motivate the limitations and future directions in Chapter 9 and the revised conclusions in Chapter 10.

# Chapter 9

# Limitations and Future Work

## 9.1 Domain Specificity

Results are measured on competitive Pokémon. Transfer to other partially observed simultaneous-action domains requires re-tokenization and new belief target definitions. The methods are domain-agnostic but empirical claims are benchmark-specific.

## 9.2 Play Strength Versus Scientific Metrics

This thesis does not evaluate live win rate against human players. Offline action match and belief accuracy answer RQ1–RQ3 as framed. They do not certify ladder competitiveness. A committee should not read the absence of live wins as a failed thesis. The intended contribution is measurement of belief learning and combinatorial generalization under offline imitation.

## 9.3 Label and Reconstruction Noise

Metamon first-person reconstruction may contain errors on edge-case mechanics. We mitigate with manual spot checks, corrupted-battle filtering, and comparison of random-bot versus human statistics.

## 9.4 Belief Labels After Reveal

Belief metrics focus on pre-reveal turns with masked slots. Including post-reveal slots would inflate accuracy trivially. Item, ability, and tera accuracies are trained but not broken out as primary tables because move slots dominate the label mass.

## 9.5 Compute and Model Scale

We intentionally use modest model size for Master's-scale hardware. Scaling width, depth, and context length may narrow gaps to large black-box sequence policies [Grigsby et al., 2025] but is left for future work.

## 9.6 Offline RL Stability

CQL and AMAGO fine-tuning can be unstable on sparse or noisy data. Stage B was not run. Strong Stage A belief and BC results, including the negative belief-to-policy transfer finding, suffice for the minimum viable bar in Chapter 6.

## 9.7 Gen 9 Team Preview

In Gen 9 OU, all opponent species are visible after team preview. Species-based belief and Split B keyed only by species composition may understate harder combinatorial shift involving unrevealed items, abilities, and move sets. Split C and finer set-combination held-out evaluation remain open (Chapter 7).

## 9.8 Future Directions

- Live evaluation on `poke-env` against `RandomPlayer`, frozen BC bots, and optionally ladder humans, reported as play-strength appendix metrics rather than as the primary thesis claim.
- Split C held-out item, ability, and move-set tuples, plus per-attribute belief tables and calibration plots.
- Sweeps over $\lambda_{\text{bel}}$ and larger models to test whether belief-to-policy transfer appears at higher capacity.
- Conservative offline RL (CQL or AMAGO) initialized from scaled BC checkpoints.
- Context-window and flat-sequence ablations for RQ4 completeness.
- Decision-tree and attention case studies on move-belief errors.
- Cross-format evaluation (other generations or tiers) for broader generalization claims.
- Release of combinatorial split manifests as a standalone benchmark contribution.

# Chapter 10

# Conclusion

Partially observed simultaneous-action games require agents to infer hidden opponent state from interaction history and act without lookahead search. This thesis studied whether a causal transformer with structured thirteen-token turns, auxiliary belief supervision, and combinatorial held-out evaluation improves offline imitation learning in that setting. Competitive Pokémon Gen 9 OU human replays via Metamon served as the benchmark. The scientific contribution is belief inference and combinatorial generalization under offline learning, not competitive ladder play.

## 10.1 Contributions

The work delivers four concrete contributions.

**Pipeline and representation.** A reproducible path from Metamon replays to encoded turn sequences, belief-enriched labels, and `poke-env` evaluation hooks. Each turn is tokenized as thirteen discrete slots with causal history over $N = 8$ prior turns.

**Model family.** A shared `TurnTransformer` trunk with incremental ablations: turn-only BC, matchup-conditioned BC (`TeamMatchupEncoder`), and `ActiveBeliefTransformer` with move, item, ability, and tera heads.

**Evaluation protocol.** Complementary splits for random battle holdout (Split A) and held-out opponent team keys (Split B), with generalization gap as the primary combinatorial statistic. A one-command scale script reproduces export, training, and OOD evaluation.

**Empirical findings on human data.** Phase 1 BC reached 18.4% action match on ≈255 000 turns (988 action classes). Phase 2 on 2 000 battles showed 20.4% random-holdout action match for turn-only BC, 28.1% move-belief accuracy for the belief-augmented model, 35.5% for a latest-turn MLP baseline, and near-zero OOD team gap after scaling. Explicit belief and matchup modules did not improve policy metrics over turn-only cloning.

## 10.2 Summary of Findings

Belief inference and policy imitation responded differently to scale. Move-belief accuracy more than doubled when the export grew from 500 to 2 000 battles. Action accuracy improved for all models but did not rank belief- augmented training above turn-only BC. Matchup conditioning over full team rosters did not help, likely because species composition is already encoded in opponent party tokens and because Gen 9 OU reveals all species at team preview.

The opponent-team generalization gap shrank from about two percentage points on the pilot corpus to under half a point at scale for every architecture. Data coverage therefore explains much of the early gap. Explicit belief and matchup modules did not outperform the trunk on Split B OOD accuracy.

Species belief, as originally envisioned for partially hidden rosters, does not apply to this format. The empirical belief results concern active-opponent move sets and related hidden attributes.

Offline imitation success should not be confused with live competitiveness. Matching human actions on roughly one turn in five leaves most decisions wrong. Against skilled human players the agent would be expected to lose almost every game. That expectation does not erase the offline scientific findings above.

## 10.3 Hypotheses Revisited

**H1** received mixed support. The transformer learns history-conditioned move beliefs well above pilot accuracy and beats a GRU baseline, but a latest-turn MLP reaches higher move-belief accuracy (35.5% versus 28.1%).

**H2** was not supported at current scale. Auxiliary belief loss did not improve action match over BC-only training, and belief fusion into the policy head also failed to help.

**H3** was mixed. Scaling reduced OOD degradation for all models. Belief and matchup modules did not achieve a smaller gap than the baseline.

**H4** (offline RL fine-tuning) remains future work.

The minimum viable bar in Chapter 6 asked for at least two of four criteria. This thesis meets the bar on (i) demonstrating nontrivial belief learning on hidden moves relative to chance and neural baselines and (iv) evidence that structured turn history and data scale contribute measurable signal, but not on belief improving OOD policy or offline RL gains. The negative result on belief-to-policy transfer is reported as a finding.

## 10.4 Broader Impact

The scientific contribution is not competitive Pokémon play. It is an empirical study of transformer belief inference and combinatorial generalization under offline imitation. The negative result on belief-to-policy transfer is informative for domains where auxiliary prediction tasks are tempting but trunk capacity may already absorb the relevant signal for action matching, or where multi-task training fails to route belief features into the policy head. Practitioners should validate whether explicit belief heads help policy metrics on their splits before adding multi-task complexity.

## 10.5 Future Work

Priority directions follow from Chapters 8 and 9.

- Live win-rate evaluation on `poke-env` as a play-strength appendix, not as a replacement for the offline claims.
- Held-out set-combination splits (Split C) and per-attribute belief tables for item, ability, and tera.
- Sweeps over $\lambda_{\text{bel}}$ and larger models to re-test belief-to-policy transfer.
- Conservative offline RL (CQL or AMAGO) initialized from scaled BC checkpoints.
- Context-window and flat-sequence ablations, plus attention or decision-tree case studies on move-belief errors.
- Larger corpus scale toward the full Metamon subset used in Phase 1.
- Release of combinatorial split manifests as a standalone benchmark artifact.

Artifacts are available under `checkpoints/`, `logs/`, and `scripts/scale_belief_research.py` for replication and extension.

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# Appendix A

# Implementation Reference

## A.1 Repository Layout

Table A.1: Key paths in the thesis codebase.

| Path | Purpose |
|---|---|
| src/pokemon_thesis/tokenizer/ | 13-token encoding |
| src/pokemon_thesis/model/ | TurnTransformer, BeliefTransformer |
| src/pokemon_thesis/data/ | Datasets and combinatorial splits |
| src/pokemon_thesis/metamon_bridge/ | Metamon state conversion |
| scripts/train_bc.py | Behavioral cloning training |
| scripts/train_belief_bc.py | Multi-task belief + action training |
| thesis/main.tex | This document |

## A.2 Building the PDF

From the repository root on Windows PowerShell:

```
cd thesis
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

Or run `scripts/compile_thesis.ps1` which executes the same sequence.

## A.3 Training Commands

Encode baseline logs:

```
python scripts/encode_logs.py logs/baseline_*.jsonl
```

Train BC TurnTransformer:

```
python scripts/train_bc.py logs/encoded_*.jsonl --epochs 30
```

Train belief-augmented model when encoded records include belief labels:

```
python scripts/train_belief_bc.py logs/encoded_*.jsonl --epochs 30
```

## A.4 TurnTransformer Forward Pass

Given token IDs `token_ids` of shape $(B, T, 13)$ and turn mask `turn_mask` of shape $(B, T)$, the model applies slot self-attention per turn, causal turn attention across history, and linear action logits over the final valid turn embedding.

BeliefTransformer adds per-slot species heads and optional fusion into the policy input when enabled.